⇒ Length Control in Kimi: (length reward):

correct answers are incentivized to be short

incorrect answers are incentivized to be shorter than the center of the range of rollouts

⇒ Rewards:

- for code: → take problems with ground truth solutions, generate new test cases.
- for math: → 800k samples to train a CoT reward model for answer equivalence checks : regex string match

Qwen3 Alibaba: Better than o1 and R1 (final case)

↓

SFT + RL Reasoning:

- ①: Filtering for difficulty (by best-of-n, n = 14 (?))
  - ⇒ remove things that the model gets right w/o CoT
  - ⇒ Remove things too similar to validation
- ②: manual filtering for the quality of (guessing right/getting it right) CoTs
- ③: RL with GRPO on only 3995 examples

↓

thinking mode fusion - controlling the length of CoT

① ⇒ Mix of non-thinking and thinking data with tags

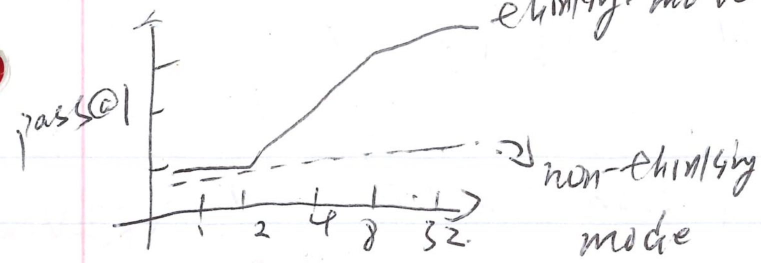
{non-thinking mode {thinking mode

⇒ Early stopping termination via a special string

thinking / no-thinking tag

<think> / <no-think>

Test-time scaling



⟸ { AIME'25/24
LiveCodeBench
GPQA Diamond }

post-training

Qwen3 overall picture:

stage 1: ; stage 2:

Base model ⟶ Long-COT cold start ⟶ Reasoning RL

flagship Models }

stage 3: Thinking Mode Fusion ⟶ stage 4: General RL ⟶ Qwen3-235B-A22B
Qwen3-32B

Lightweight model:

Base models ⟶ strong-to-weak-distillation ⟶ Qwen3-30B
14B/8B/4B

Qwen3: post-training pipelines

~~SFT: over-fitted / biased~~

pretrain: so model speak language / gen token

↓

SFT : ~~test~~ follow instruction: → over-fitted / biased

↓

RLHF : Behave like human prefer → ~~alig~~ alignment / safety

but: noisy / sparse Rewards
Rewards hacking
subjective

↓

RLVR : { Be correct when human fails
replaces human judgement with objective signals → Binary / no-noise
Not subjective / cheat